Swynex Internship – Task 2

Data Cleaning and Basic Analysis

1. Objective

The objective of this task is to retrieve a biological nucleotide sequence, clean the sequence data, and perform basic sequence analysis.

2. Dataset Information

- Organism: *Homo sapiens*
- Gene: Insulin (INS)
- Accession Number: NM_000207.3
- Sequence Type: mRNA
- Source: NCBI Nucleotide Database

3. Methodology

The nucleotide sequence was retrieved from NCBI using Biopython and Entrez. The sequence was converted to uppercase and whitespace was removed to prepare the data for analysis. The sequence length, nucleotide counts, and GC content were calculated using Python.

4. Results

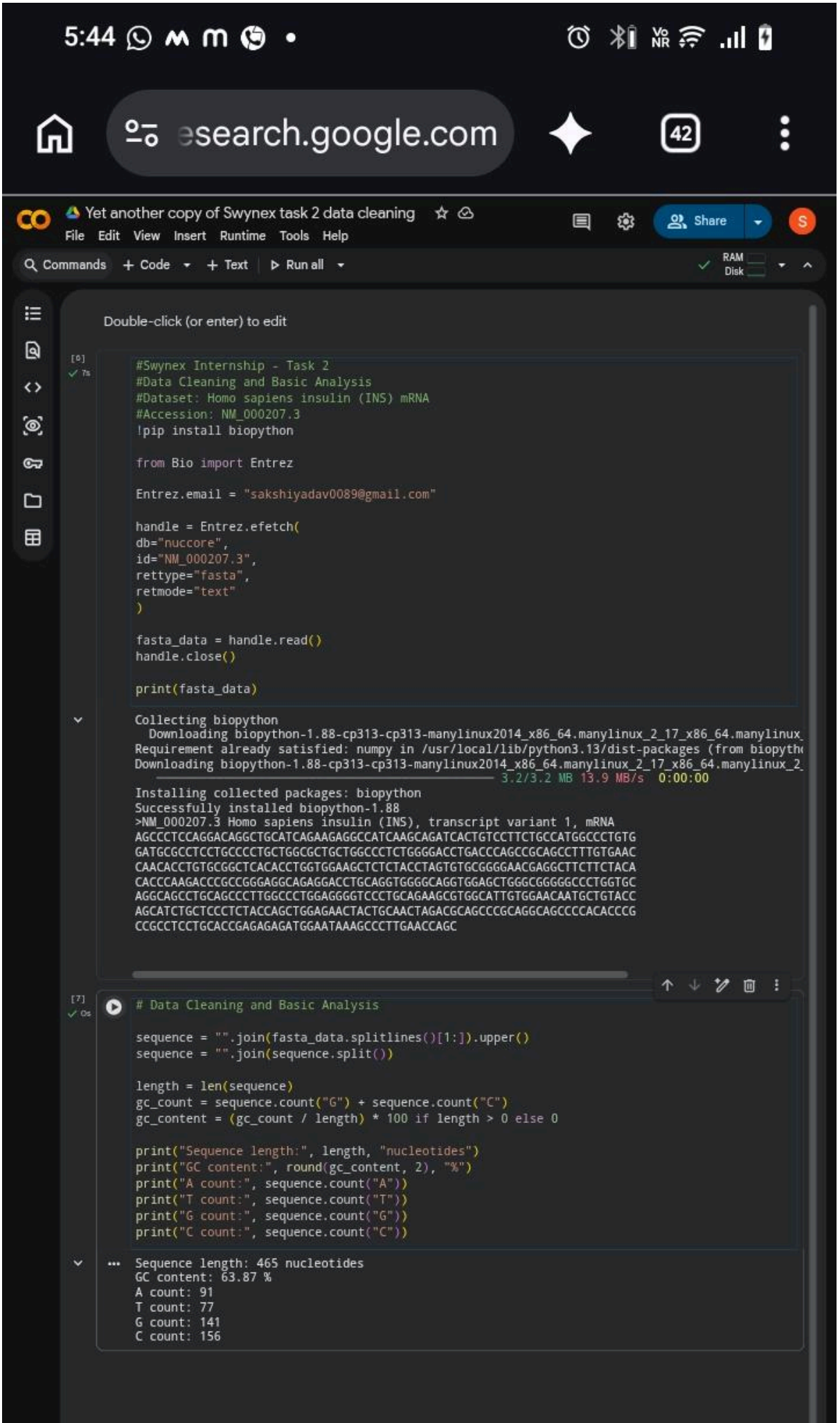
- Sequence length: 465 nucleotides
- Adenine (A) count: 91
- Thymine (T) count: 77
- Guanine (G) count: 141
- Cytosine (C) count: 156
- GC content: 63.87%

5. Conclusion

This task provided practical experience in retrieving biological sequence data, cleaning nucleotide sequences, and performing basic analysis using Python. The results helped demonstrate how computational tools can be used to analyse biological data.

6. Evidence

A screenshot of the Python code and its output is included as evidence of the analysis performed.


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Yet another copy of Swynex task 2 data cleaning
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Double-click (or enter) to edit
[6]
7s
#Swynex Internship - Task 2
#Data Cleaning and Basic Analysis
#Dataset: Homo sapiens insulin (INS) mRNA
#Accession: NM_000207.3
!pip install biopython
from Bio import Entrez
Entrez.email = "sakshiyadav0089@gmail.com"
handle = Entrez.efetch(
db="nuccore",
id="NM_000207.3",
rettype="fasta",
retmode="text"
)
fasta_data = handle.read()
handle.close()
print(fasta_data)
Collecting biopython
Downloading biopython-1.88-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (from biopyth
Downloading biopython-1.88-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_
3.2/3.2 MB 13.9 MB/s 0:00:00
Installing collected packages: biopython
Successfully installed biopython-1.88
>NM_000207.3 Homo sapiens insulin (INS), transcript variant 1, mRNA
AGCCCTCCAGGACAGGCTGCATCAGAAGAGGCCATCAAGCAGATCACTGTCCTTCTGCCATGGCCCTGTG
GATGCGCCTCCTGCCCCTGCTGGCGCTGCTGGCCCTCTGGGGACCTGACCCAGCCGCAGCCTTTGTGAAC
CAACACCTGTGCGGCTCACACCTGGTGGAAGCTCTCTACCTAGTGTGCGGGGAACGAGGCTTCTTCTACA
CACCCAAGACCCGCCGGGAGGCAGAGGACCTGCAGGTGGGGCAGGTGGAGCTGGGCGGGGGCCCTGGTGC
AGGCAGCCTGCAGCCCTTGGCCCTGGAGGGGTCCCTGCAGAAGCGTGGCATTGTGGAACAATGCTGTACC
AGCATCTGCTCCCTCTACCAGCTGGAGAACTACTGCAACTAGACGCAGCCCGCAGGCAGCCCCACACCCG
CCGCCTCCTGCACCGAGAGAGATGGAATAAAGCCCTTGAACCAGC
[7]
0s
Data Cleaning and Basic Analysis
sequence = "".join(fasta_data.splitlines()[1:]).upper()
sequence = "".join(sequence.split())
length = len(sequence)
gc_count = sequence.count("G") + sequence.count("C")
gc_content = (gc_count / length) * 100 if length > 0 else 0
print("Sequence length:", length, "nucleotides")
print("GC content:", round(gc_content, 2), "%")
print("A count:", sequence.count("A"))
print("T count:", sequence.count("T"))
print("G count:", sequence.count("G"))
print("C count:", sequence.count("C"))
Sequence length: 465 nucleotides
GC content: 63.87 %
A count: 91
T count: 77
G count: 141
C count: 156